

CLASS XI · LECTURE 3

The System Unit, Registers and the Fetch Cycle

Two periods · Long-question material: registers, buses, fetch cycle

By the end of this lecture you can

- Name the parts of the CPU and the job of ALU and CU.
- Define the main registers and draw the fetch cycle.
- Distinguish address bus, data bus and control bus.
- Name ports a student actually sees on a system unit.

What is inside the system unit

The system unit is the box that holds the working parts. Opening it is a practical period: you should be able to point at the power supply, the motherboard, the processor, the RAM modules, the hard drive, the optical drive if there is one, the cooling fan, and the ports on the back. The exam asks for the jobs, not for a repair guide.

Part	Job
Power supply	Converts mains electricity into the low voltages the boards need
Motherboard	The main circuit board. Every other part plugs into it or is soldered on it
Microprocessor	The CPU. Runs the program
RAM modules	Main memory used while a program runs
Hard drive	Secondary storage
Cooling fan and heat sink	Carry heat away from the processor
Ports	Sockets for keyboard, mouse, printer, display and USB devices

The CPU**DEFINITION Central processing unit**

The CPU is the part that fetches instructions from memory, decodes them and carries them out. It has two main sections: the arithmetic and logic unit and the control unit, plus a small set of registers.

The **ALU** adds, subtracts, multiplies, divides and compares. A comparison is a logical test, such as “is this mark at least 40?”. The **control unit** does not calculate. It directs. It fetches the next instruction, decodes it, and opens the right paths so that data moves among memory, ALU and input/output at the right moment. If the ALU is the calculator, the control unit is the person who reads the question and presses the right keys.

Clock speed, measured in gigahertz, is the number of timing pulses the processor receives each second. A faster clock can start more steps per second. It is not, by itself, the whole story of speed: the number of cores, the cache and the memory also matter. You only need the definition unless a question asks for factors.

Registers

A register is a small, very fast storage location inside the CPU. It holds one piece of data, one address or one instruction for the step that is happening now. A long question, “What is a computer register? Explain its types,” is answered from this table. Write the definition, then five registers with one line each.

Register	Holds	Why it exists
Program counter (PC)	Address of the next instruction	So the CPU knows where to fetch from

Register	Holds	Why it exists
Instruction register (IR)	The instruction just fetched	The control unit decodes what is in the IR
Memory address register (MAR)	The address of a memory location about to be used	It is placed on the address bus
Memory buffer register (MBR or MDR)	The data or instruction moving to or from memory	It is the CPU's side of a memory transfer
Accumulator (AC)	The current arithmetic result	The ALU works into the accumulator
General-purpose registers	Temporary values for the running program	They save trips to main memory

The fetch cycle

An instruction cycle is fetch, then decode, then execute. The **fetch cycle** is only the first part: getting the next instruction into the instruction register. Questions that say “draw the fetch cycle” want the steps below, in order. Do not skip the program counter.

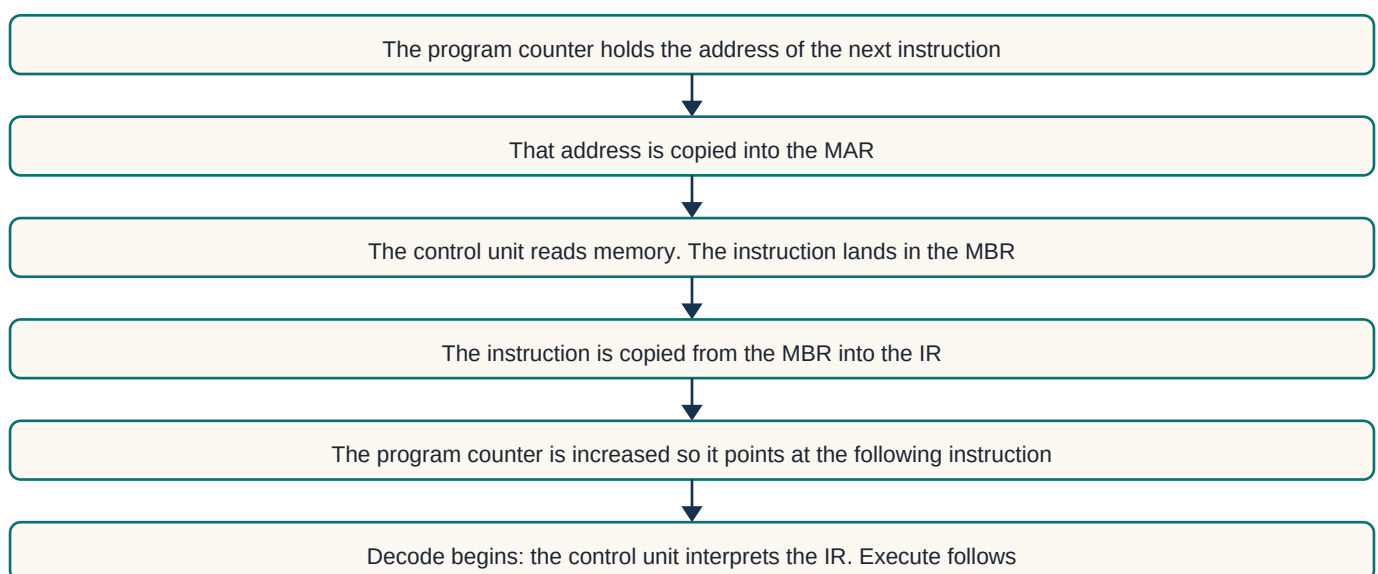


Figure 3.1 Fetch cycle. Decode and execute are the rest of the instruction cycle.

EXAM TIP A diagram that scores

Five boxes are enough: PC, MAR, Memory, MBR, IR. Arrow from PC to MAR labelled “copy address”. Arrow from Memory to MBR labelled “read instruction”. Arrow from MBR to IR. A small note beside the PC: “then PC = PC + 1”. Label the figure. An unlabelled circle does not score.

Buses

DEFINITION Bus

A bus is a shared electrical path that carries data, addresses or control signals from one part of the computer to another.

Bus	Carries	Direction
Address bus	The address of a memory location or a port	One way: from the CPU out to memory
Data bus	The actual data or instruction	Both ways: read from memory, or write back
Control bus	Signals such as read, write and clock	Both ways, depending on the signal

A wider data bus moves more bits at once. A 64-bit data bus can carry 64 bits in one transfer. The address bus decides how much memory can be named. These two widths are the “bus width” a specification sheet

talks about. Cache is a small fast memory beside the CPU that keeps copies of data the processor is likely to need again, so it does not wait for RAM every time.

The motherboard around the CPU

You should recognise these parts if a practical or a short question names them. **BIOS** is the Basic Input Output System, a program in ROM that checks the machine at power-on and starts loading the operating system. The **CMOS battery** keeps the date, time and setup when the mains power is off. **RAM slots** hold the memory modules. **Expansion slots** (PCI and the older EISA) take extra cards such as a display card or a network card. **Jumpers** are small links that older boards used to select a setting. Ports face the outside of the case.

Port	Character	Typical use
Serial	One bit at a time	Older mice and some instruments
Parallel	A group of bits together	Older printers
USB	A general serial bus, hot-pluggable	Printers, phones, pen drives, keyboards
HDMI	Picture and sound together	Monitors and projectors
VGA	Older analogue video	Older projectors and CRT monitors

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The electrical path that carries signals from one computer part to another is a:

- A) Bus
- B) Port
- C) Plotter
- D) Virus

2. The register that holds the address of the next instruction is the:

- A) Accumulator
- B) Instruction register
- C) Program counter
- D) MBR

Short answers

S1. What is the fetch cycle? Draw its diagram.

S2. Differentiate the address bus and the data bus.

Detailed answer

L1. What is a computer register? Explain its types.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. A. A bus is the shared path. A port is the socket on the outside.

MCQ 2. C. The IR holds the current instruction. The PC holds the address of the next one.

S1. The fetch cycle brings the next instruction from memory into the CPU. The program counter's address is copied to the MAR. Memory is read and the instruction arrives in the MBR, then moves into the IR. The program counter is then increased. Draw PC to MAR to Memory to MBR to IR, and mark $PC = PC + 1$.

S2. The address bus carries the address of the memory location the CPU wants to use, and it runs from the CPU towards memory. The data bus carries the data or instruction itself, and it runs in both directions: into the CPU on a read, and out of the CPU on a write.

L1. A register is a small, fast storage location inside the CPU. It holds the address, instruction or value needed for the current step. The program counter holds the address of the next instruction. The instruction register holds the instruction that has just been fetched, so the control unit can decode it. The memory address register holds the address that will be placed on the address bus. The memory buffer register holds the data or instruction moving between the CPU and memory. The accumulator holds the result of the ALU. General-purpose registers hold temporary values for the program so that every value does not have to be written back to RAM at once. Registers are faster than RAM and much smaller.